LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 6

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO6

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 6

Unit Title: Evaluation and Applications of Unsupervised Learning

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the external validation measure that corrects the Rand Index for the agreement expected by chance.

A) The Adjusted Rand Index
B) The silhouette coefficient
C) The Davies-Bouldin Index
D) The inertia of the partition

Q2. Identify the two properties that a good clustering has to combine.

A) Accuracy and precision
B) Cohesion and separation
C) Recall and specificity
D) Support and confidence

Q3. State the silhouette value conventionally assigned to a record that is the only member of its cluster.

A) 1
B) -1
C) 0
D) It is undefined

Q4. Name the internal index built from the ratio of between-cluster dispersion to within-cluster dispersion.

A) The Adjusted Rand Index
B) The Jaccard index
C) The confidence of a rule
D) The Calinski-Harabasz index

Q5. Identify what a cluster profile reports.

A) The distance between every pair of records
B) The typical feature values of each cluster
C) The number of iterations the run took
D) The random seed used for the run

Q6. State the value the Adjusted Rand Index takes when two partitions agree completely.

A) 0
B) -1
C) The number of clusters
D) 1

Q7. Explain why an external validation measure cannot be applied to a wholly unlabelled dataset.

A) It compares the clusters against reference labels
B) It requires the features to be standardised first
C) It can only be computed for exactly two clusters
D) It needs the algorithm to be run several times

Q8. Distinguish cohesion from separation in a clustering.

A) Separation is closeness within, cohesion is distance between

B) Both of the two describe closeness within a cluster
C) Cohesion is closeness within, separation is distance between
D) Both of the two describe distance between clusters

Q9. Describe what consensus clustering sets out to do.

A) Run one algorithm until the inertia is zero
B) Compare a clustering against known class labels
C) Combine many runs into one agreed partition
D) Reduce the features before the clustering starts

Q10. Interpret an Adjusted Rand Index close to zero between a clustering and a set of reference labels.

A) The agreement is no better than chance
B) The two partitions agree almost perfectly
C) The clustering has exactly zero clusters
D) The reference labels were recorded wrongly

Q11. Predict what happens to a customer segmentation over a year as buying behaviour shifts.

A) It becomes more accurate as data accumulates
B) It remains valid because it uses no labels
C) It automatically updates itself from new records
D) It drifts out of date and has to be rebuilt

Q12. Recognize the ethical risk described: a segmentation avoids using ethnicity but relies on postcode, browsing language and preferred store.

A) The features are too few for a reliable result
B) Proxy features can reintroduce a protected attribute
C) The features are all measured on the same scale
D) The features make the clustering run too slowly

Q13. Explain why the sizes of the clusters should always be reported with a clustering result.

A) They are needed to compute the silhouette score
B) They determine the random seed that was used
C) They fix the number of features to retain
D) They expose a degenerate partition that scores well

Q14. Describe why the silhouette score should be reported together with the number of clusters chosen.

A) It states how many iterations the run required
B) It states which features were used in the run
C) It states how well supported that particular choice is
D) It states the proportion of records left as noise

Q15. Explain why an unsupervised model still needs monitoring once it is deployed.

A) The algorithm gradually loses its parameters
B) The data it describes keeps changing over time
C) The features become correlated after deployment
D) The absence of labels makes it unstable to run

Q16. Distinguish purity from the Adjusted Rand Index as external measures.

A) Purity ignores chance agreement and rewards many clusters
B) Purity corrects for chance while the Rand Index does not
C) Both of the two correct for the agreement due to chance
D) Neither of the two requires any reference labels

Q17. Interpret a clustering that has placed 98 per cent of the records into a single cluster.

A) The data holds one extremely strong group
B) The result is degenerate and must be reconsidered
C) The silhouette score must therefore be zero
D) The algorithm converged unusually quickly

Q18. Predict the effect on interpretability of clustering on principal components rather than on the original features.

A) The segments become much harder to describe
B) The segments become much easier to describe
C) Interpretability is unaffected by the change
D) The segments can no longer be given labels

Q19. Explain why transparency matters when customers are segmented on their behaviour.

A) Segmentation is forbidden without written consent
B) Transparency improves the silhouette of the result
C) Transparency reduces the cost of running the model
D) Decisions are being made about people who cannot see them

Q20. Recognize the case study described: images of handwritten digits are projected into two dimensions to see whether the ten digits separate visually.

A) An association rule mining of purchase baskets
B) An anomaly detection study of network traffic
C) A dimensionality-reduction visualisation of MNIST
D) A hierarchical clustering of customer records

Q21. Apply the silhouette definition to a record whose mean distance to its own cluster is 4 and to the nearest other cluster is 10.

A) 0.6
B) 0.4
C) 1.5

D) 0.24

Q22. Compute the purity of a cluster holding 50 records in which the most common true label appears 40 times.

A) 0.4
B) 1.25
C) 0.8
D) 0.2

Q23. Use the Adjusted Rand Index to compare two clusterings and determine what a value of 1 signifies.

A) The two partitions share no record at all
B) The two partitions are identical up to renaming
C) The two partitions each hold exactly one cluster
D) The index has been computed incorrectly

Q24. Solve the following: a segmentation is rebuilt on 10 bootstrap samples and one particular segment reappears in 9 of them. Determine its stability score.

A) 0.1
B) 9.0
C) 0.09
D) 0.9

Q25. Apply an external validation measure and determine what it requires before it can be computed.

A) A second run of the same algorithm
B) A reduction of the data to two dimensions
C) Known labels for at least a subset of records
D) An equal number of records in every cluster

Q26. Examine a clustering that reports a high silhouette score but whose three clusters hold 995, 3 and 2 records, and determine what to report.

A) The partition is excellent and should be used
B) The silhouette score must be recomputed
C) The number of clusters should be increased
D) The partition is degenerate despite the score

Q27. Compare internal validation with external validation on what each of the two needs.

A) Internal needs geometry alone, external needs labels
B) External needs geometry alone, internal needs labels
C) Both of the two need a set of reference labels
D) Both of the two need only the geometry of the data

Q28. Use cohesion and separation to determine which of two partitions of the same data is the better.

A) The one with the larger number of clusters
B) The one with tighter clusters that are further apart
C) The one with the smaller number of clusters
D) The one produced by the faster algorithm

Q29. Compute the number of record pairs the Rand Index has to consider for a dataset of 10 records.

A) 100
B) 90
C) 10
D) 45

Q30. Apply a stability check to a clustering and determine the correct procedure.

A) Run the algorithm once with a fixed random seed
B) Re-cluster many subsamples and measure the agreement
C) Increase k until the inertia stops falling
D) Compare the result against the training labels

Q31. Use a cluster profile to give a segment a business name.

A) Use the cluster number the algorithm assigned
B) Choose the record nearest the cluster centre
C) Compare its feature averages with the overall averages
D) Count how many records the cluster contains

Q32. Examine a segmentation whose groups differ on one feature alone and determine the first thing to check.

A) Whether the features were placed on a common scale
B) Whether the algorithm converged successfully
C) Whether the number of records was sufficient
D) Whether the random seed had been recorded

Q33. Compare Principal Component Analysis with t-SNE for the panels of a visualisation dashboard.

A) PCA gives reproducible axes, t-SNE clearer local groups
B) t-SNE gives reproducible axes, PCA clearer local groups
C) Both of the two give axes that can be interpreted
D) Neither of the two is suitable for a dashboard

Q34. Apply the unsupervised workflow and determine the stage at which the segments are given their business names.

A) Before any of the features are chosen
B) While the distance measure is selected
C) During the standardisation of the columns
D) After the clusters have been profiled

Q35. Solve the following: a record has a mean distance of 6 to its own cluster and 6 to the nearest other cluster. Determine its silhouette value.

A) 1
B) 0
C) -1
D) 0.5

Q36. Use the Jaccard index to measure how far a segment survives a resample when 40 records are shared and 60 records appear in at least one version.

A) 0.4
B) 1.5
C) 0.667
D) 0.24

Q37. Examine an insurer that clusters customers on postcode and behaviour and then offers different premiums by cluster, and determine the principal risk.

A) The clustering will converge too slowly to use
B) The silhouette score will be too low to report
C) Indirect discrimination through a proxy variable
D) The number of clusters will be hard to choose

Q38. Demonstrate how a completed clustering should be handed over to a business team.

A) With the cluster labels and nothing further
B) With profiles, sizes, validation figures and stated limits
C) With the full distance matrix of the records
D) With the source code that produced the result

Q39. Relate the interval at which a segmentation is rebuilt to the rate at which the behaviour it describes changes.

A) Faster-changing behaviour calls for fewer rebuilds
B) The interval is fixed once the model is deployed
C) The interval depends only on the number of records
D) Faster-changing behaviour calls for more frequent rebuilds

Q40. Use two clusterings of the same data to determine what an external measure reports when one has simply had its cluster labels renamed.

A) Perfect agreement, since the groupings are identical
B) No agreement, since the labels are all different
C) Partial agreement, in proportion to the renaming
D) The measure cannot be computed in this case

----- End of Sample Question Set -----